neutralize the three-phase line, forms a shunt path through the length of the line. Therefore, capacitance effects introduce complications in transmission line calculations.

The overhead transmission lines are classified as

i. Short transmission lines
ii. Medium transmission lines
iii. Long transmission lines

i. Short transmission lines

When the length of an overhead transmission line is about 50 metres and line voltage about 20 KV is usually called as a short transmission line. Due to smaller length and low voltage, the capacitance effects are small. The total resistance and inductance are assumed to be at one point for calculation purpose.

ii. Medium transmission lines

When the length of an overhead transmission line is to a span of 50 m to 150 m and line voltage 20 KV to 100 KV. It is usually called as a medium transmission line. Due to sufficient length and voltage of the line, the capacitance effects are taken into account. For the purpose of calculation, the distributed capacitance of the line is divided in the form of condensers shunted across the line at one or more points.

iii. Long transmission lines

The length of an overhead transmission line is more than 150 metres and line voltage is 100 KV, it is called as a long transmission line.

1.8.1 Guarding

Low, medium and high voltage conductors are caused by natural disasters, such as rain or storm which cause electrical accidents. This accident causes damage to life. If a live conductor is cut down, the earthed cradle part is used to hold the conductor and protects lives. This part is called as guarding. When the conductor touches the guarding, the circuit breaker automatically disconnects the supply.

Types of guarding

i. Cradle guarding
ii. Cage guarding

1.9 Line insulators

The overhead line conductors should be supported with the poles or towers by means of insulators. These insulators act as supports in order to avoid any leakage of current from the conductor to earth.

1.9.1 Properties of insulators

Here are some of the properties of line insulators

i. Mechanical strength should be very high.
ii. Its dielectric strength should be very high.
iii. Insulators must be free from internal defects such as impurities to leakage current.
iv. Electrical insulation value of resistance must be high.
v. Environmental conditions should not be affected.
vi. Do not have porous.
vii. Price should be cheaper.

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